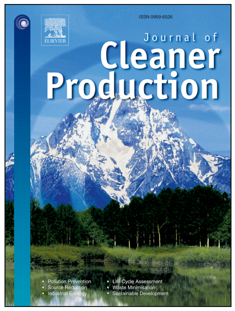

Methanol production via direct carbon dioxide hydrogenation using hydrogen from photocatalytic water splitting: Process development and techno-economic analysis

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PII: S0959-6526(18)33148-2

DOI: 10.1016/j.jclepro.2018.10.132

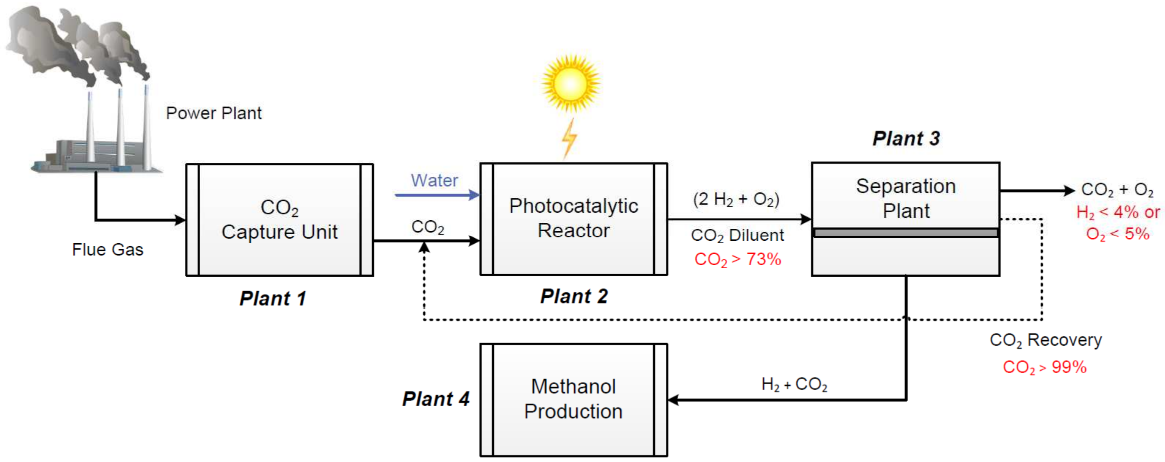
Reference: JCLP 14534

To appear in: *Journal of Cleaner Production*

Received Date: 26 March 2018

Revised Date: 25 September 2018

Accepted Date: 11 October 2018



Power Plant
Flue Gas
CO_2 Capture Unit
Plant 1
Water
CO_2
Photocatalytic Reactor
Plant 2
$(2 H_2 + O_2)$
CO_2 Diluent
CO_2 > 73%
Plant 3
Separation Plant
$CO_2 + O_2$
H_2 < 4% or
O_2 < 5%
CO_2 Recovery
CO_2 > 99%
$H_2 + CO_2$
Plant 4
Methanol Production

Word count in Microsoft document = 8091 words (including references)

Methanol Production via Direct Carbon Dioxide Hydrogenation using Hydrogen from Photocatalytic Water Splitting: Process Development and Techno-Economic Analysis

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Abstract

In this work, we demonstrate the process and economic viability of a new production route for methanol utilizing captured CO_2 and H_2 produced from renewable, photocatalytic water splitting. Initial assessments of H_2 production from this approach revealed a low-cost potential through inexpensive feedstock, but with one major drawback. Simultaneous production of H_2 and O_2 increases the risk of forming an explosive mixture. In our previous work, we identified membrane-based processes to safely recover the H_2 product. To reduce the H_2 capture costs, we developed a conceptual process comprised of integrated facilities to produce methanol while solving the flammability constraints. The first unit supplies captured CO_2, which is used as a raw material and a diluent to recover H_2 from the photocatalytic reactors. The tertiary mixture ($H_2/O_2/CO_2$) is safely processed in an optimized membrane-based separation plant to produce a 3:1 H_2 and CO_2 mixture. This binary mixture is then used as feedstock for methanol production via direct CO_2 hydrogenation. Based on a detailed economic analysis, the break-even value of the methanol produced is higher than the current market price. The main drivers for the high cost are the separation plant and the feedstock costs of CO_2. The best-case scenario (for the selected parameters), where the plant lifetime is 30 years with 0.060 \$/kWh electricity price and 0.020 \$/kg CO_2 cost, has a break-even value for methanol at 0.96 \$/kg. Based on a target solar-to-hydrogen efficiency of 10% in the photocatalytic reactor, the gross energy efficiency of the process was 6.25%. As a final element, we compared our process with alternative renewable methanol synthesis route from the literature to highlight the differences in electrochemical and thermochemical approaches.

Keywords: Methanol production, carbon dioxide utilization, membrane separation, water splitting, solar to chemicals, optimization

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1. Introduction

Accounting for more than 60%, CO_2 emissions from burning fossil fuels is the main contributor between other greenhouse gases for global warming [1, 2]. Recent studies and observations show the potential harmful impacts of these gases on our environment including rising sea levels and superstorms [3, 4, 5]. Industrial nations have to implement global efforts to reduce CO_2 emissions and limit the temperature increase to 2°C [2, 6]. One approach to mitigate this problem is to utilize Carbon dioxide Capture and Reuse (CCR) processes for fuels and value-added chemical production, which can partially close the carbon cycle in the atmosphere [7, 8]. Besides the reduction of CO_2 emissions, this approach would also generate economic benefits and positive social-political images [9].

Methanol is one of the major chemical products that can be produced using captured CO_2, which has a potential market of more than 80 million tons per year globally [10]. It is a key building block for numerous commodity chemicals (i.e. MTBE, acetic acid, formaldehyde) [11]. As a liquid fuel, methanol is considered a suitable energy storage component due to its high energy density by volume and weight, and easy handling and storage compared to H_2 [9, 12, 13]. Conventionally, methanol is produced from a mixture of CO, H_2 and CO_2 commonly known as synthesis gas or syngas. Most of the world's syngas is generated from natural gas via steam reforming or partial oxidation (auto-thermal reforming) [14, 15]. Countries rich in coal feedstocks, such as China and South Africa, use coal gasification systems to generate syngas as well [16]. This step in the conventional methanol synthesis accounts for more than 60% of the capital investment and requires substantial amounts of water and energy, around 35 GJ/ton$_{MeOH}$, which is also provided from fossil fuels [17, 18]. The syngas gas mixture is converted to methanol over a metal-based catalyst (e.g. $CuO/ZnO/Al_2O_3$) catalyst according to the following reactions [11]:

$$CO + 2\,H_2 \longrightarrow CH_3OH$$

$$CO_2 + 3\,H_2 \longrightarrow CH_3OH + H_2O$$

A typical natural gas based methanol plant will produce 0.5 t_{CO_2}/t_{MeOH} [17]. To achieve an overall CO_2 reduction while synthesizing methanol, a renewable source of energy and/or feedstocks must be utilized [8, 19].

Solar energy has the potential to provide a sustainable syngas source for the methanol production using captured CO_2 and water based feedstocks. One method is water splitting wherein H_2 is produced and then partially utilized to produce CO through a reverse water gas shift (RWGS) reaction using captured CO_2. Alternatively, the produced H_2 can be utilized for the direct hydrogenation of CO_2 to produce methanol [9, 17, 20]. A second method is using solar energy to reduce CO_2 directly to form CO in a catalytic system. After CO formation, conventional water gas shift (WGS) is performed to produce H_2 from water and CO. These two approaches can be accomplished by either thermochemical or electrochemical conversions via solar energy.

Using the thermochemical approach to activate CO_2 to form CO, Kim, *et al.* [19, 21] reported a full techno-economic analysis of this approach. The findings of this study show that the required break-even price for the methanol is \$1.22/kg$_{MeOH}$ compared to the current price of \$0.38/kg$_{MeOH}$ [22]. The high production cost is driven by the solar concentrator/reactor, which accounts for more than 90% of the capital investment. Similarly, the same technique can be used to generate H_2 from water. Abanades and Flamant [12, 23] utilized a CeO_2/CeO_3 catalyst to split water into H_2 and O_2 using a concentrated solar reactor. Other catalytic materials are also being utilized for this purpose (e.g. Fe_3O_4/FeO and ZnO/Zn) [12].

For the electrochemical approach (including photocatalysis and photoelectrolysis), suitable semiconductor materials generate electrons and holes via solar energy, which are used to react with water and CO_2. The main obstacle facing this approach is the dissolution of CO_2 in aqueous solution to enable the reduction to CO. Elevated pressures are needed to overcome limited CO_2 solubility (33mM at 25°C and 1 atm), which adds considerable processing costs [5]. Other challenges include formation of byproducts in the aqueous solution, challenging separations, and catalyst deactivation. The other feedstock, H_2, can also be produced from water. Photoelectrochemical (PEC) water splitting to produce H_2 and O_2 has attracted a lot of attention from the scientific community for more than 40 years. The first paper described this approach was presented by Fujishima and Honda in 1972 using TiO_2 photoelectrode [24, 25]. Recently, a full techno-economic analysis of four different technologies utilizing this approach for water splitting has been reported by the US Department of Energy (DOE) [26, 27]. The findings of this study showed that the single-bed colloidal suspension reactor, which utilizes photocatalytic nanoparticles, has the lowest estimated H_2 production price of \$1.6/kg_{H_2}. This process has its challenges as well. Both H_2 and O_2 are produced in the reactor without any physical separation, which drastically increases the risk of explosion [12]. The DOE study discounted the explosion limits, which makes the process industrially unrealistic. In our previous work [28, 29], we explored the utilization of different diluents, namely N_2 and CO_2, to recover the produced H_2 while maintaining rigorous flammability constraints throughout the separation process. We were able to produce a high purity H_2 product (99 mol%) and achieve agreeable recovery rates. However, the resulting economics show imposing recovery costs due to low H_2 concentration in the feed.

Building upon our previous work [28, 29], we have identified an opportunity to utilize renewable PEC water splitting as a source of H_2 coupled with CCR to create a new approach for methanol production (see Fig.1). Leveraging the flexibility of membrane separations, we have designed a process scheme that enables on-purpose production of the necessary ratio of CO_2 and H_2 that can be utilized in a conceptual CO_2 hydrogenation reactor. A CO_2 capture unit (Plant 1) will supply pure CO_2, as a raw material and diluent, to the photocatalytic reactor (Plant 2) which safely transfers the reaction products. The resulting ternary mixture is sent to a membrane separation plant (Plant 3) to recover both H_2 and CO_2 with a preferred stoichiometric ratio of 3:1 (H_2:CO_2). Residual CO_2 from Plant 3 is recovered to minimize costs and energetic footprint. The H_2/CO_2 mixture is compressed and fed to the methanol loop in Plant 4 (included product recovery units). Our focus was to minimize the Present Value (PV) of all outgoing cash flows of Plant 3, as this plant represents the new approach for the methanol production in this study. Stringent flammability constraints were applied during the optimization process to ensure safe operating conditions. Optimum or ideal case studies regarding Plants 1, 2, and 4 were extracted from the literature and were assimilated into the overall economic and energetic analyses. To the best of our knowledge, this is the first fundamental process design

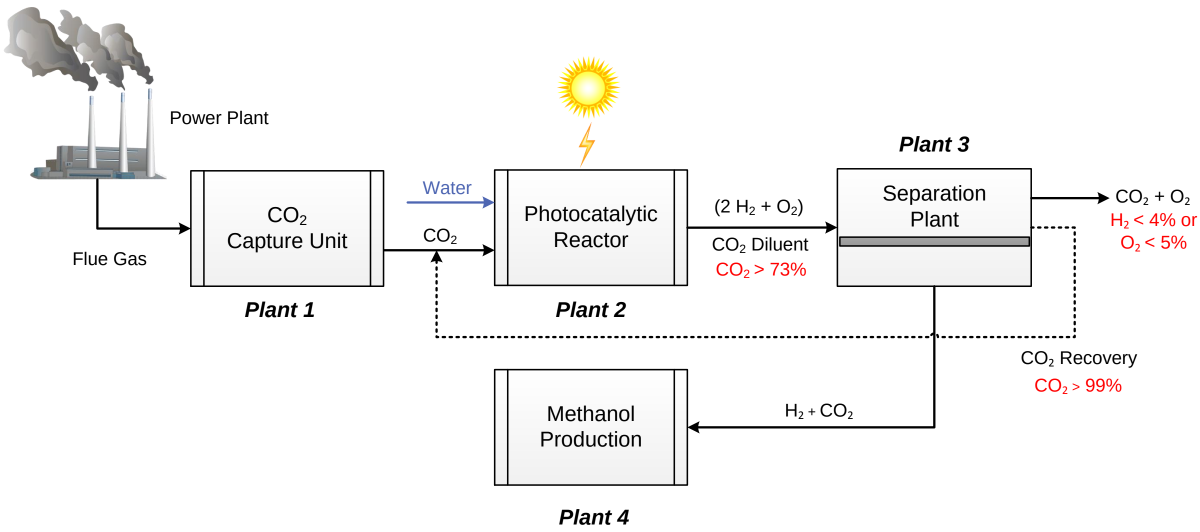


Figure 1: General layout of the proposed methanol synthesis from photocatalytic water splitting and captured CO_2.

for methanol synthesis using H_2 from water splitting and captured CO_2 as both a diluent and a raw material.

2. Process description and economics

The process design aims to utilize a renewable source and method of H_2 production from photocatalytic water splitting (as estimated in the study of US DOE [26]) while mitigating the flammability risks inherent to the simultaneous production of H_2 and O_2. Moreover, the separation plant (Plant 3) is designed to recover H_2 with lower purity than our previous objective [29] which will result in a cost reduction for the proposed process, comparatively. These design elements are combined with ideal scenarios reported for the other plants in the overall process concept. The following general assumptions concerning the four plants were taken into consideration in this work:

- Each plant will be designed individually with no heat integration.
- The process will be designed based on a H_2 production rate of 1000 kmol/h from Plant 2.
- The cost of feed from Plants 1 and 2 will be treated as purchasing raw materials.
- A pure CO_2 stream will be supplied by Plant 1.
- In Plant 3, the targeted purity of CO_2 recovery is 99 mol%, which will be recycled back to Plant 2 with no additional treatment.
- Trace amounts of O_2 leaving Plant 3 will be converted to H_2O using a stoichiometric reactor (O_2 removal unit) in Plant 4. The minor cost of this unit will be neglected in this study.

In the following subsections, we will provide a brief description of Plants 1 and 2 along with the primary parameters used for both economic and energy analysis. The separation plant (Plant 3) will be based on our previous work

[29]. However, new membrane layouts design along with updated economic parameters and constraints (for CO_2 recovery) will be implemented in the optimization problem. Finally, Plant 4 will be adopted from the work of Van-Dal and Bouallou [20] but with an alternate feed stream composition based on the surrogate raw materials sources. The process economics will be calculated using Guthrie's method with an estimated error of ±25% [30, 31]. Moreover, sensitivity analysis on both economic and energy parameters of the process will be carried out, namely:

- CO_2 capture price ($\$/kg_{CO_2 produced}$)
- Electricity price ($/kWh)
- Plant lifetime (yr)
- Energy requirement for CO_2 capture ($kWh/kg_{CO_2 processed}$)
- Solar-to-Hydrogen (STH) efficiency (%)

2.1. Plant 1: CO_2 capture

Both cost and energy consumption for CO_2 capture from power plants are strongly dependent on the fossil fuel type (e.g., oil, coal, natural gas, biomass) and the technology used to produce the electricity or heat [6]. Some examples of power plant technologies include: Supercritical Pulverized Coal (SCPC), Natural Gas Combined Cycle (NGCC), and Integrated Gasification Combined Cycle (IGCC) [32]. There are numerous proposed technologies for large scale CO_2 capture, such as: liquid amine-based absorption, solid-state adsorbents (pressure or thermal swing) [33], membranes [34], and hybrid concepts [35]. Amine based solvents, however, are the most prevalent and mature solution for CO_2 capture [36]. Carbon dioxide capture technologies are integrated within fossil fuel power plants via one of three primary designs [6, 32, 37, 38]:

- *Post-combustion Capture:* CO_2 is removed from the flue gas after complete combustion of fossil fuels with air. The low-pressure flue gas contains around 16 mol% of CO_2.
- *Pre-combustion Capture:* CO_2 is removed before the combustion of H_2, which is produced via steam reforming of the fossil fuel to produce syngas. The potential stream for CO_2 capture contains around 38 mol% CO_2 at elevated pressures ($\approx 30 bar$).
- *Oxyfuel Combustion:* Concentrated O_2 is utilized instead of air to burn the fossil fuel. Doing so removes the necessity to separate diluted CO_2 from N_2 [39]. A high-quality CO_2 stream is generated after removing water vapor.

For this study, the base case will use standard parameters to calculate the cost of CO_2 as a raw material along with energy consumption. These parameters and the respective values used in this work are provided in Table 1.

Table 1: Process and economic parameters for the proposed process

	Base Case		
	Parameter	Value	Unit
Process Parameters	CO_2 capture efficiency [1, 39]	90.0	%
	CO_2 energy requirement [1]	0.30	$kWh/kg_{\mathrm{CO_2 processed}}$
	STH efficiency [26]	10.0	%
	Catalyst density[20]	1775	kg_{cat}/m^3_{cat}
	fixed bed porosity[20]	0.4	-
	Catalyst lifetime [40]	2	yr
	catalyst wt. to CO_2 feed	0.50	$kg_{cat}/(kg_{\mathrm{CO_2}}/h)$
Economic Parameters	CO_2 capture price [1, 32]	0.035	$\$/kg_{\mathrm{CO_2 produced}}$
	H_2 production price[26, 27]	0.65	$\$/kg_{\mathrm{H_2 produced}}$
	Annual operating hours	8000	h
	Plant lifetime (T)	15	yr
	Interest rate (i)	9.0	%
	Currency exchange [41]	1.1	€ to \$
	Electricity price [42]	0.087	$\$/kWh$
	Water price [17]	0.033	$\$/m^3$
	Membrane price [43]	500.0	$\$/m^2$
	Membrane replacement [43]	100.0	$\$/m^2.yr$
	Update Factor*	4.75	-
	Catalyst price [44]	100.0	$€/kg_{cat}$
	Sensitivity Analysis		
	Parameter	Range	Unit
Energy Parameters	CO_2 energy requirement	0.15-0.45	$kWh/kg_{\mathrm{CO_2 processed}}$
	STH efficiency	5.0-15.0	%
Economic Parameters	CO_2 capture price [1, 32]	0.020-0.035	$\$/kg_{\mathrm{CO_2 produced}}$
	Plant lifetime (T)	15-30	yr
	Electricity price	0.060-0.087	$\$/kWh$

** Chemical Engineering Plant Cost Index (Nov. 2016)*

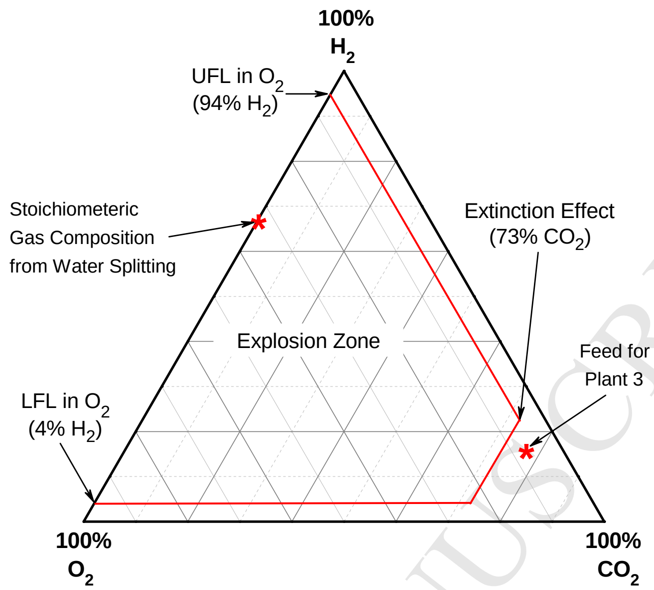


Figure 2: Hydrogen flammability diagrams in $H_2/O_2/CO_2$ mixture under normal operating conditions [29].

2.2. Plant 2: Photocatalytic reactor

Photoelectrochemical water splitting is initiated when solar photons are absorbed by a semiconductor material. Hence, a transparent slurry reactor, made of high-density polyethylene (HDPE), is used to allow sun light to enter the reactor. The semiconductor nano-sized catalysts are suspended in a potassium hydroxide (KOH) electrolyte solution. An example catalyst is 40 nm iron oxide (Fe_2O_3) with a 5nm shell layer of TiO_2 [27]. Upon reacting with sunlight, the catalysts produce electron-hole pairs that react with water to drive both the H_2 and O_2 evolution reactions. The benchmark value of solar-to-hydrogen (STH) efficiency is 10% [26]. Produced gases from this reactor are sent to a product recovery facility (Plant 3) to yield H_2. Previous reports describe the use of a pressure swing adsorption (PSA) design to recover H_2 [26]. Upon further evaluation of the cost of the production analysis [26, 27], we see that more than 60% of the capital cost and a majority of the operating costs are associated with gas compression and purification. Furthermore, the separation step, utilized in these previous reports [26, 27], was estimated under the assumption of no explosion limitation between H_2 and O_2 even though the produced stoichiometric gas composition (66.7 mol% H_2 and 33.3 mol% O_2) is obviously within the explosive region as shown in Fig. 2.

In our study, CO_2 received from Plant 1 will be utilized to dilute and sweep the produced stoichiometric gas to a membrane-based separation facility (Plant 3). The composition of the diluted stream entering Plant 3 is shown in Table 2. Base case parameters for Plant 2 are shown in Table 1 including the updated estimated production price of H_2 (60% lower than previous reports [26]). Additionally, a range of parameter values for the corresponding sensitivity analysis are also listed in Table 1.

Table 2: Plant 3 feed compositions for the proposed process

Gas	Concentration
Carbon dioxide (H_2)	76 mol%
Hydrogen (H_2)	16 mol%
Oxygen (O_2)	8 mol%

2.3. Plant 3: Separation plant

Membranes have the potential to compete with other conventional separation processes (e.g. PSA) to separate the tertiary mixture of $H_2/O_2/CO_2$ leaving Plant 2. Membrane Technology and Research, Inc. (MTR), has commercialized two membranes of interest applicable to our process: H_2-selective (Proteus™) and CO_2-selective (Polaris™) membranes. These membranes are produced using a spiral-wound module design and have also been investigated for pre-combustion CO_2 capture applications [43, 45, 46]. Table 3 shows the gas permeation properties of these two membrane materials. Permeation data are reported at 30°C for Polaris™and at 150°C for Proteus™.

Table 3: Gas permeation properties of Polaris™and Proteus™[43, 46, 47]

Component	Proteus™ Permeance (GPU)	Polaris™ Permeance (GPU)
H_2	300	100
O_2	6	50
CO_2	20	1000

Fig.3 shows two membrane process layouts which will be optimized to recover both H_2 and CO_2, at a ratio of 3:1, while minimizing the O_2 concentration in the outlet stream entering the methanol synthesis facility (Plant 4). Layout (a) has no recycle stream; however, it utilizes two vacuum pumps in membrane units M1 and M2 on the permeate. Layout (b), on the other hand, requires only one vacuum pump for the M1 permeate. The retentate from M2 is recycled back to the first membrane unit (M1) which increases the driving force and enhances the process recovery. In both configurations, the M3 membrane unit is comprised of a CO_2-selective membrane to purify and recover residual CO_2 to 99 mol%, which will be sent back to the Plant 2 sweep stream thereby minimizing the release of additional CO_2.

The optimization problem is solved by nonlinear programming (NLP) to minimize the objective function which is the Present Value (PV) of all outgoing (negative) cash flows including the cost of raw materials (CO_2 from Plant 1 and H_2 from Plant 2). Using this approach will achieve the optimization objectives along with minimizing the loss of raw materials. General Algebraic Modeling System (GAMS) is used to solve the NLP based on a global optimization solver (BARON). Table 4 shows the optimization equations for the proposed separation plant (Plant 3).

Although not shown, both process layouts illustrated in Fig.3 include the necessary compressors, vacuum pumps and heat exchangers to manipulate the decision variables, namely: recycle rates, membrane area, and feed/permeate

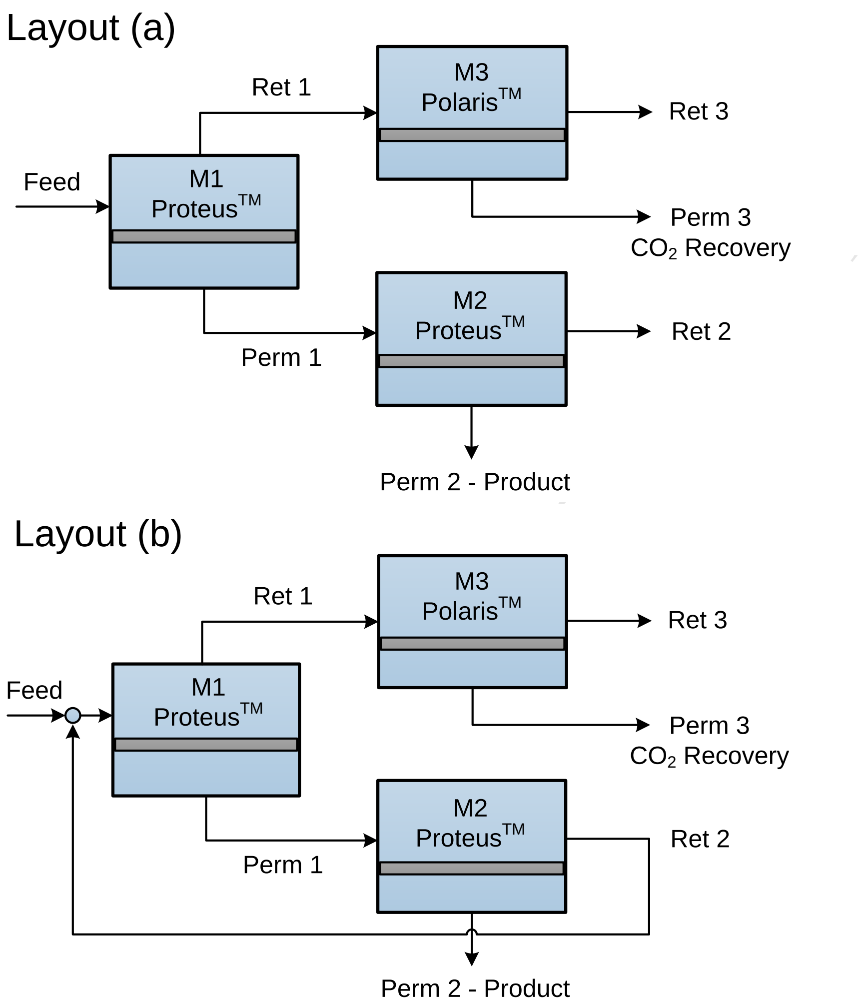


Figure 3: Two membrane process layouts for Plant 3.

pressure in order to achieve the results. All details of these process units, including the membrane units, were described in our previous work [28, 29]. Table 1 shows the updated economic parameters used to calculate the PV and the sensitivity analysis.

2.4. Plant 4: Methanol production

The conversion of CO_2 into renewable methanol can be achieved via two different approaches. A one-step approach, also known as direct CO_2 hydrogenation, utilizes a renewable H_2 source and captured CO_2 to directly form methanol over a Cu-based catalyst [9, 20]. The two-step approach requires the formation of syngas from CO_2 (via renewable energy source) which is then converted to methanol through a conventional process [20]. In this paper, the one-step approach to produce methanol was employed.

Early investigations were carried on the use of Cu-based catalysts for the conversion of CO_2 to methanol [38, 48, 49, 50]. More specifically, the commercially available catalyst for methanol production from syngas ($Cu/ZnO/Al_2O_3$)

Table 4: NLP model for Plant 3

Equation	Definition	Values
Objective function: f(x)	Total cost of all outgoing cash flows (no income) of the process including: investment & operating cost, raw materials and depreciation for n years.	minimize
Constraints g(x)⩾0	Sets of linear and nonlinear process constraints including:	
	- H_2 recovery	⩾ 80%
	- H_2 purity in Perm 2 stream	= 75%
	- CO_2 purity in Perm 3 stream	⩾ 99%
	- H_2 limits in retentate streams	⩽ 4.0%
	- Max. streams pressure (absolute)	⩽ 30 bar
	- Min. permeate pressure (absolute)	⩾ 0.3 bar
	- Max. trans-membrane pressure (TMP_{max}) [46]	⩽ 12 bar
Model equations: h(x)=0	Sets of the equipment equations required to calculate the performance and economics	-

can be used for CO_2 hydrogenation according to the following two reactions:

$$CO_2 + 3\,H_2 \longrightarrow CH_3OH + H_2O$$

$$CO_2 + H_2 \longleftrightarrow CO + H_2O$$

Reaction kinetics were adopted from the work of Van-Dal and Bouallou [20] which are suitable for direct implementation into Aspen Plus using the LHHW (Langmuir-Hinshelwood-Hougen-Watson) reaction model. The reactions will be carried out in an adiabatic packed bed reactor using the parameters listed in Table 1. A detailed process flow sheet is described in the Results and Discussion section.

2.5. Overall process economic and energy analysis

As stated before, the economic calculations for the Capital Cost (CC) will be estimated using Guthrie's method. This method utilizes the bare module cost (BC) equation which is then corrected by three different factors: Update Factor (UF), Material and Pressure Factor (MPF), and Module Factor (MF) as shown in the following equations [31]:

$$BC = C_0 \cdot (\tfrac{S}{S_0})^{\alpha} \tag{1}$$

$$CC = UF \cdot BC \cdot (MPF + MF - 1) \tag{2}$$

For the sake of brevity, all equipment specifications and parameters will be summarized in the supporting information. After the CC and Operating Cost (OC) calculations, the PV and the NPV can be calculated as follows [51]:

$$PV_{i,T} = CC + OC \cdot \frac{(i+1)^T - 1}{(i+1)^T \cdot i} \tag{3}$$

$$NPV_{i,T} = R_t \cdot \frac{(i+1)^T - 1}{(i+1)^T \cdot i} - CC \tag{4}$$

where R_t is the net cash flow resulting from the annual difference between the methanol sales and operating cost of the proposed process. The total NPV will be calculated using the current methanol price of 0.38/kgMeOH [22] and the base case parameters listed in Table 1.

The Specific Cost (SC) can be calculated by dividing the Equivalent Annual Cost (EAC) by the annual production rate:

$$EAC = \frac{PV}{A_{T,i}} \tag{5}$$

$$A_{T,i} = \frac{1 - \frac{1}{(1+i)^T}}{i} \tag{6}$$

$$SC = \frac{EAC}{Annual\ production\ rate} \tag{7}$$

All four plants will be compared through the cash flow required to process the base case flow rate of 1000 kmol/h from Plant 2. For this purpose, the following specific costs are calculated:

- *SC(1)* = CO_2 capture price ($\$/kg_{CO_2 \mathrm{produced}}$)
- *SC(2)* = H_2 production price ($\$/kg_{H_2 \mathrm{produced}}$)
- *SC(3)* = Specific cost of Plant 3 ($\$/kg_{H_2 \mathrm{recovered}}$)
- *SC(1-3)* = Specific cost of Plant 3 including raw materials ($\$/kg_{H_2 \mathrm{recovered}}$)
- *SC(4)* = Specific cost of Plant 4 ($\$/kg_{\mathrm{MeOH\ produced}}$)

Two energy efficiencies will be utilized to compare with other solar energy based methanol production [19]:

$$Net\ Energy\ Efficiency = \frac{MeOH\ chemical\ energy}{\mathrm{H}_2\ chemical\ energy\ +\ Utilities}$$

$$Gross\ Energy\ Efficiency = \frac{MeOH\ chemical\ energy}{Total\ solar\ energy\ +\ Utilities}$$

The net energy efficiency is the ratio of the chemical energy of the produced methanol over the chemical energy of the produced H_2 from Plant 2 via sunlight (solar energy) and the utilities. The gross energy efficiency accounts for the total solar energy required for H_2 production from Plant 2. The chemical energy for methanol and H_2 will be calculated using the Higher Heating Value (HHV) of 726 MJ/kmol$_{MeOH}$, and 286 MJ/kmol$_{H_2}$, respectively.

3. Results and discussions

As mentioned before, the derivative costs from Plants 1 & 2 will be estimated using the prices of CO_2 capture and H_2 production. To obtain the mixture compositions listed in Table 2 for the base case, an initial supply of 4750 kmol/h for CO_2 from Plant 1 is required for plant start-up. However, the net amount of CO_2 consumption will be determined after the optimization study of Plant 3 is completed in order to estimate the CO_2 recovery.

3.1. Plant 3: Separation plant

3.1.1. Optimization results

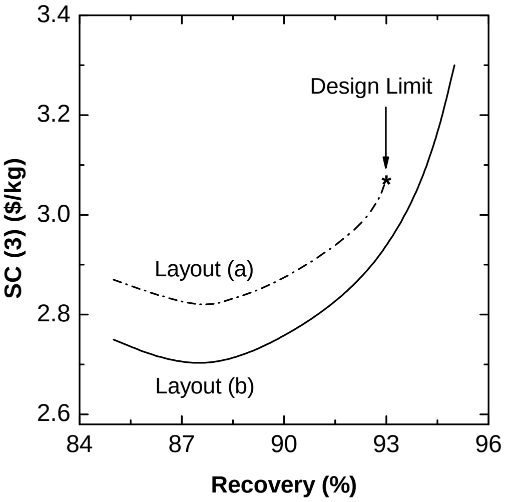
The optimization aims at a H_2 purity of 75 mol% with a minimum recovery of 80%. Optimizing the process to minimize the separation costs without considering raw materials cost led to misleading results by ignoring the CO_2 recovery membrane unit (M3). Thus, it was essential to optimize the process to minimize SC (1-3) instead of SC (3) for both layouts. The optimization pronlem for layout (a) could not be solved when all constraints listed in Table 4 were implemented. This was resolved by reducing the CO_2 purity in Perm 3 stream to 98.8% for this layout only. Moreover, layout (a) suffered from a limited recovery rate of 93% for H_2. This design limit is related to the maximum trans-membrane pressure (TMP_{max}) of 12 bar as shown in Fig.4.

Fig.4a shows clearly the lower cost of operating layout (b) over layout (a), even with all constraints implemented. The optimum recovery for the separation process should be around 87.5% based on the SC (3) analysis. However, there is a trade-off. At 87.5% H_2 recovery, the CO_2 recovery is limited (¡ 15%), as shown in Fig.4b, which drives the overall specific costs (SC 1-3) higher when raw materials are considered. Based on Fig.4c, both layouts exhibit similar cost at high H_2 recovery rate with a distinct advantage for layout (b) for having higher CO_2 purity in the recovery stream and higher H_2 recovery rates. Layout (a) would be attractive if lower H_2 recovery rates were acceptable, but that is unlikely due to the intrinsic economic value of H_2. Fig.5 and Table 5 show the optimum results for layout (b), which will be used as the basis of the complete investigation for this study.

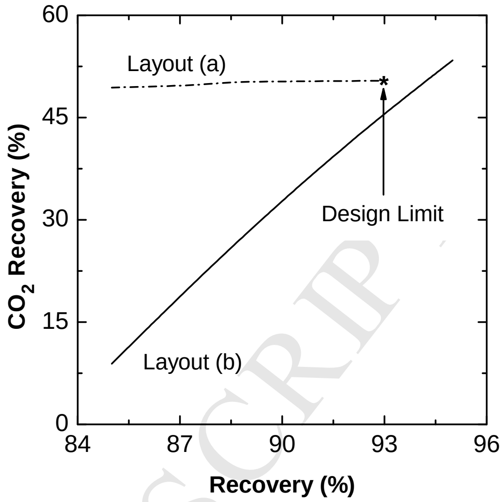
As shown in Table 5, there are trace amounts (ppm) of O_2 in the product stream entering the methanol reactor (Plant 4). There are no reports in the open literature describing the effects of O_2 on the commercial methanol process or catalyst; which is logical as the conventional syngas composition does not contain O_2 [52, 53]. Rather than ignore the residual O_2, we considered three different scenarios to address this challenge:

- *Scenario (1)*: Treat O_2 as an inert gas in Plant 4
- *Scenario (2)*: O_2 reaction with H_2 to form water inside the methanol reactor in Plant 4
- *Scenario (3)*: O_2 removal unit with negligible cost

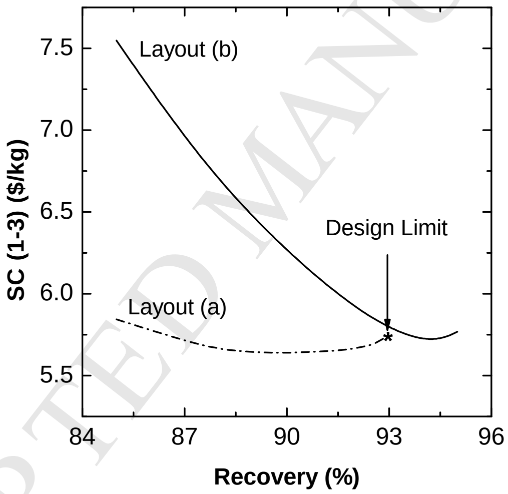
Scenario (1) was not considered since O_2 is likely to be very active on the copper based catalysts. Additionally, the Aspen Plus simulation showed a slight increase in product loss in Plant 4 due to requisite increase in the flare flow rate to prevent O_2 build up in the reaction loop. The challenge for Scenario (2) was to find reaction kinetics (LHHW)



(a) Separation specific cost SC (3) profile



(b) CO_2 recovery rate profile



(c) Separation specific cost SC (1-3) profile

Figure 4: Optimization results for Plant 3.

that were suitable for Aspen Plus to combine with direct CO_2 hydrogenation kinetics (none have been reported to date). Therefore, Scenario (3) was selected for this study. Some processes were described in the literature concerning the removal of residual O_2 in other gas H_2 containing mixtures similar to our study [54, 55] and there are even commercially available technologies (e.g. X−O_2™from Newpoint Gas, Ltd.).

Table 5: Optimum process and economic results for Plant 3 (base case)

Parameter	Value	Unit
H_2 recovery (Perm 2)	94.2	%
CO_2 recovery (Perm 3)	50.3	%
O_2 (Perm 2)	2480	ppm
Capital Cost (CC)	119.72	Million $
Operating Cost (OC)	32.05	Million $/yr
Raw materials cost	39.57	Million $/yr
SC (3)	3.08	$\$/kg_{H_2\,\mathrm{recovered}}$
SC (1-3)	5.69	$\$/kg_{H_2\,\mathrm{recovered}}$

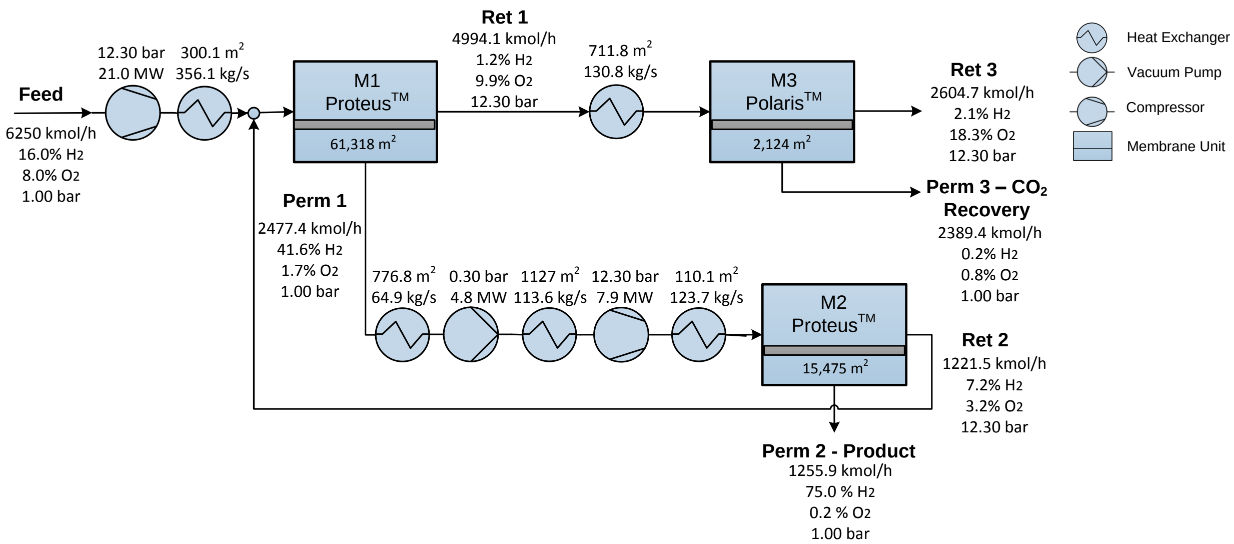
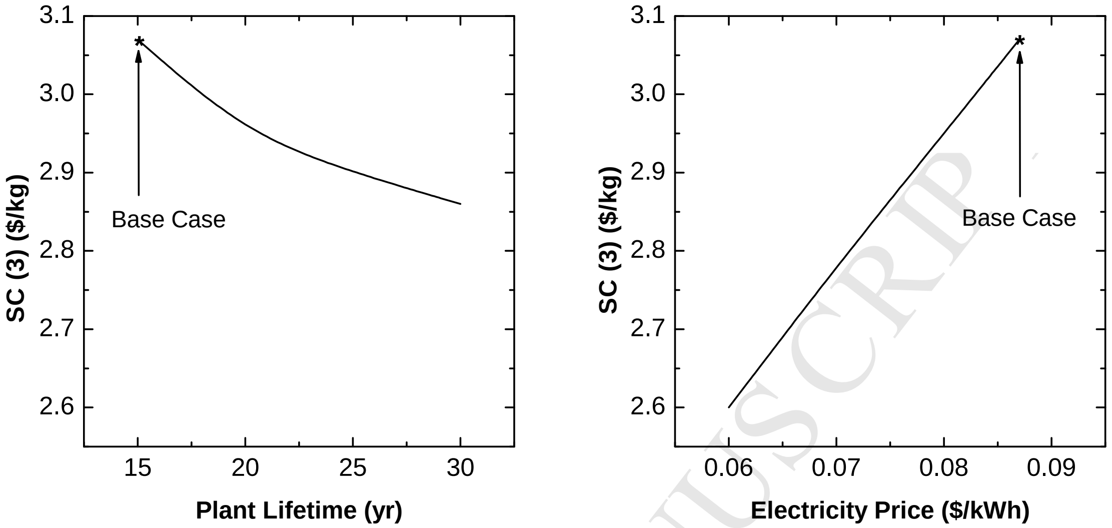


Figure 5: Optimum process layout for Plant 3.

3.1.2. *Sensitivity analysis*

The results shown in Table 5 are obtained using the base case parameters. Two of these parameters were chosen to be part of the sensitivity analysis affecting Plant (3) cost, namely: plant lifetime and electricity price. Fig.6 summarizes the findings where specific cost SC (3) is reduced by 6.8% when the plant lifetime increases from 15 to 30 years of operation. More influence can be noticed when the electricity price decreases from 0.087 to 0.06 $/kWh, where the specific cost drops linearly by 15.3%. This reduction is a direct result from our previous findings [29] where the electricity cost accounts for approximately 70% of the total operating costs to run the membrane separation plant.



(a) Plant lifetime influence at 0.087 $/kWh (b) Electricity price influence at 15 years old plant

Figure 6: Sensitivity analysis for Plant 3 (SC (3)).

3.2. Plant 4: Methanol production

3.2.1. Flowsheet description

The process flowsheet for the methanol synthesis loop is shown in Fig.7 and was adopted from the work of Van-Dal and Bouallou [20]. Fresh feed is delivered to the gate from Plant 3 and is comprised of H_2, CO_2, and trace amounts of O_2. The O_2 is converted to water using a catalytic conversion reactor with the following reaction stoichiometry:

$$2\,H_2 + O_2 \longrightarrow 2\,H_2O$$

This conversion results in a minor loss of H_2, about 0.66%, from the main feed entering Plant 4. Four adiabatic compressors (isotropic efficiency of 0.80) with intermediate cooling (cooling water) are utilized to compress the feed to 75bar. The compressed feed (Stream 9) is mixed with the recycle stream (Stream 22) from the reactor and preheated (HX-5) to 210°C using a fraction (60%) of the reactor outlet (Stream 13). The preheated feed (Stream 11) enters a fixed bed adiabatic reactor filled with 6840 kg of Cu/ZnO/Al_2O_3 catalyst. The calculated reactor volume is 6.33 m^3. The reactor products are divided into two streams (Streams 13 & 14) at an elevated temperature of 284°C. Stream 14 is used as an integrated heat source for the reboiler in the distillation column (D-1) and preheater (HX-7). The divided streams are recombined and cooled via HX-6 to 35°C. The first knock-out drum (Drum-1) separates the unreacted gases (Stream 19) from methanol/water liquid products (Stream 20). A fraction (1%) of Stream 19 is purged to prevent the accumulation of by-products. The remaining unreacted gasses (Stream 22) are recycled and combined with Stream 9 before returning to the reactor.

Liquid Stream 20, also known as crude methanol, is expanded to 1.2 bar (VLV-1) which separates most of the

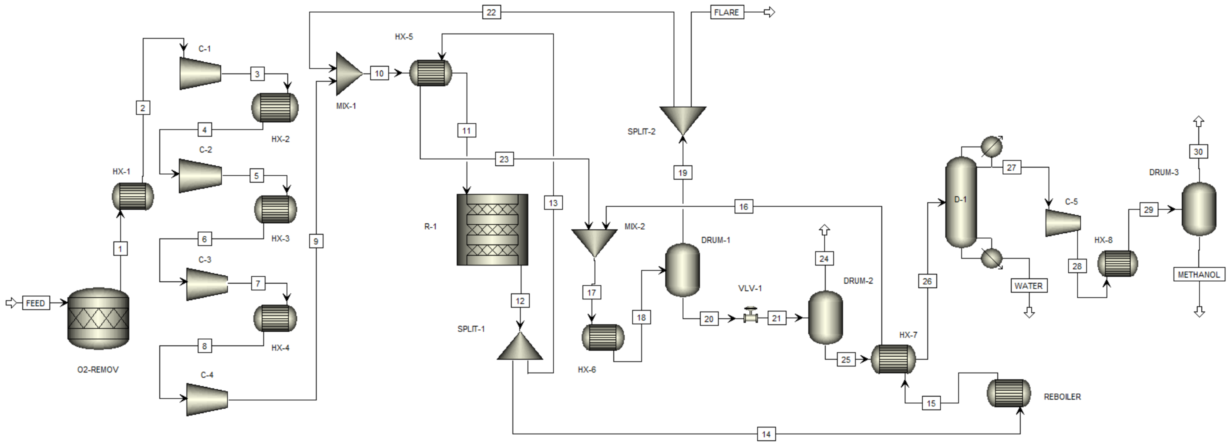


Figure 7: Process flowsheet for Plant 4.

unreacted gasses from the liquid product in the second flash drum (Drum-2). The liquid stream leaving the second drum is preheated (HX-7) to 80°C before entering the distillation column (D-1). The column is operated with a reflux ratio of 1.1 over 33 trays. The bottom stream (102°C and 1.1 bar) is water with 171 wt-ppb of methanol. The distillate stream (64°C and 1.0 bar) contains the purified methanol along with 58 wt-ppm of water and some unreacted gases. This stream (Stream 27) is compressed and cooled to 40°C as a final polishing step to remove the unreacted gases in the third flash drum (Drum-3). Detailed stream information is available in the supporting information.

3.2.2. Economic results

Plant 4 capacity is designed for a production rate of 73,056 t_{MeOH}/yr (≈ 73 kta) delivering a methanol purity of 99.37 wt%. Table 6 summarizes the process and economic results for Plant 4 based on the base case parameters. Analysis of the capital and operating costs shows that a majority of the costs (75.8% and 84.1%, respectively) are related to investment and operation cost of the compressors (a similar trend was observed for Plant 3).

Table 6: Process and Economic results for Plant 4 (base case)

Parameter	Value	Unit
Methanol production	9132.0	kg/h
Methanol yields	0.67	kg_{MeOH} / kg_{CO_2}
Methanol purity	99.37	%wt
Capital Cost (CC)	28.29	Million $
- Heat exchangers	20.6	%
- Compressors	75.8	%
- Flash drums	0.9	%
- Distillation column	2.2	%
- Reactor	0.5	%
Operating Cost (OC)	4.82	Million $/yr
- Cooling water	8.1	%
- Electricity	84.1	%
- Catalyst replacement	7.8	%
SC (4)	0.114	$\$/kg_{\text{MeOH produced}}$

3.3. Overall process

3.3.1. Economic and energy analysis

Looking at the overall process economics, the total capital cost required for Plants 3 & 4 is $148 million with an annual operating cost of 76.44 million $/yr (including the raw materials from Plants 1 & 2). Using the current methanol price of \$0.38/kg_{MeOH}, our proposed process results in a negative NPV of $540.40 million over a 15-year plant lifetime. A methanol price of \$1.30/kg_{MeOH} is the required value for the break-even point in our base case; which is more than three times higher than the current methanol price produced from natural gas. Building on the base case parameters and the 1000 kmol/h H_2 production from Plant 2, Fig.8 represents the negative cash flow required for each plant. These values were calculated given the purchase price of CO_2 and H_2 from Plants 1 & 2, respectively, and the calculated specific cost for the separation (SC 3) and methanol (SC 4) plants. Around 49% of the overall operating cost is associated with Plant 3 which is directly linked to the recovery of diluted H_2 at a high feed flow rate. The process constraints and requirements result in a high demand for membrane area and energy to deliver the desired product. Purchasing the CO_2 comes in second with more than 30% of the cost.

The energy analysis calculation is summarized in Table 7 and detailed calculations are provided in the supporting information. Non-solar energy required for the photocatalytic reactor (Plant 2) was neglected since the gas processing part of the DOE study was replaced by Plant 3 in our work. The low value of the gross energy efficiency (6.25%) was expected as the efficiency of solar-to-hydrogen (STH) for water splitting is relatively low compared to other techniques, like photovoltaic cells. Both efficiencies could be improved by optimizing the heat integration between

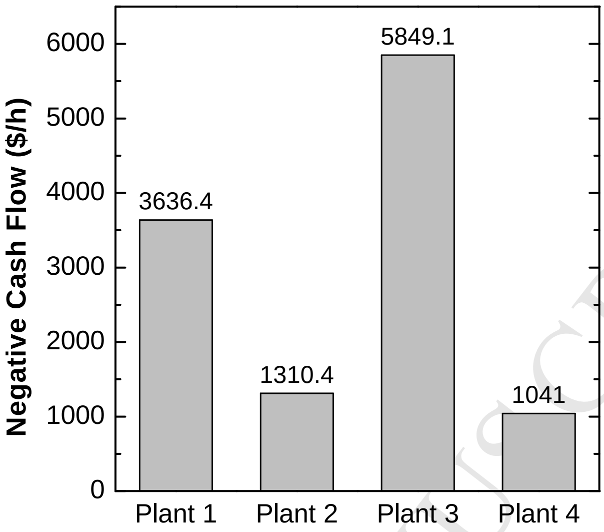


Figure 8: Negative cash flow analysis for the overall process economic (base case).

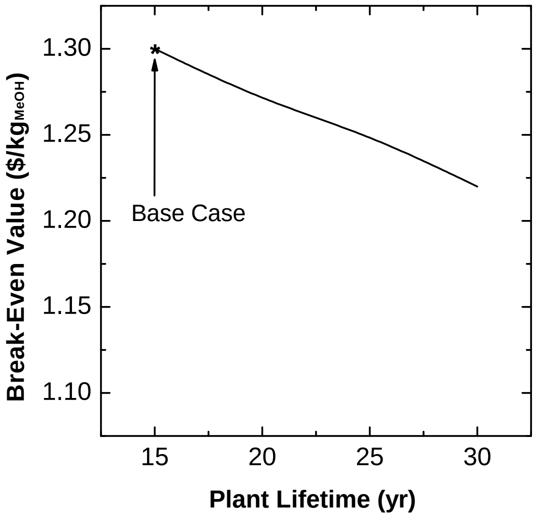
Table 7: Energy analysis for the overall process (base case)

Energy consumption	MWh/t_{MeOH}	MMBtu/t_{MeOH}
MeOH chemical energy	6.29	21.46
H_2 chemical energy	8.70	29.69
Total solar energy	87.00	296.86
Plant 1 energy	3.79	12.93
Plant 2 utility	Negligible	Negligible
Plant 3 utility	7.32	24.98
Plant 4 utility	2.51	8.54
Net Energy Efficiency	28.18 %	
Gross Energy Efficiency	6.25 %	

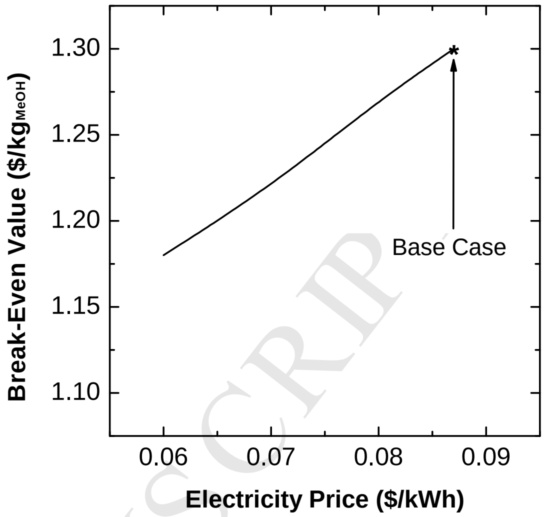
the four plants and utilizing higher performance membrane materials in Plant 3, which will reduce the compressor duties due to lower recycle rates.

3.3.2. Sensitivity analysis

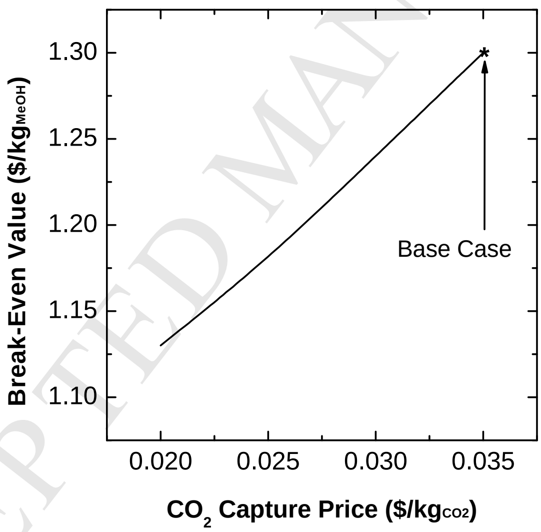
The process sensitivity with respect to the primary economic parameters (plant lifetime, electricity price, and CO_2 capture cost) is represented by calculating the break-even value for the methanol price as shown in Fig.9. Plant lifetime has the weakest influence since it only shows a 6.2% reduction in the break-even value. The cost of the CO_2 raw materials has the greatest impact (13.1% reduction) when the capture price drops from 0.035 to 0.020 $/kg$_{CO_2}$. The best case scenario, where the plant is operated for 30 years with 0.060 $/kWh electricity price and 0.020 $/kg$_{CO_2}$, has a break-even value for methanol at 0.96 $/kg$_{MeOH}$, which is around 26.5% reduction from the base case result, but



(a) Plant lifetime influence



(b) Electricity price influence



(c) CO_2 capture price influence

Figure 9: Process economic parameters influence on break-even value of methanol.

still 2.5 times more than the current price of fossil fuel derived methanol.

The energy required for the CO_2 capture in Plant 1 was varied to highlight its influence on the process energy efficiency. Surprisingly, it has a minor influence on the net energy efficiency (±2.5%) and negligible influence on the gross energy efficiency (±0.1%) as shown in Table 8. The gross energy is dominated by the total energy required for the hydrogen production, which was manipulated by different values of STH. The efficiency value increases by 2.5% when the STH value reaches 15%.

Table 8: Energy sensitivity analysis for the overall process

Parameter	CO_2 capture energy (kWh/kg$_{CO_2\ processed}$)			STH (%)		
	0.15	0.30	0.45	5	10	15
Net Energy Efficiency (%)	30.80	28.15	25.98	Not applicable		
Gross Energy Efficiency (%)	6.37	6.25	6.14	3.35	6.25	8.78

3.3.3. Comparison with other alternative processes

As explained previously, solar energy can be utilized in different ways to produce methanol using captured CO_2. Kim and coworkers built an exemplary case study for such process based on a solar conversion technology to produce CO (and H_2 via WGS) from CO_2 and H_2O [19]. We have compared our findings with their work. As a special case for this comparison, we recalculated our results using similar economic parameters as shown in Table 9.

The thermochemical approach utilizes more CO_2, converted to CO, to produce the same amount of methanol. Obviously, an extraordinary amount of CO is required to carry out both water gas shift (to produce H_2) and conventional methanol synthesis. Whereas, the direct hydrogenation process proposed in our study required less CO_2 as a raw material feed. Regarding the total process energy, both studies promote similar net energy efficiencies, with CO_2 hydrogenation having a 16% edge. The thermochemical approach, on the other hand, is more efficient (65%) in terms of gross energy. The Solar-to-Chemical efficiency is at 20% for the thermochemical approach, which is two times higher than what is assumed for the STH in our work. Finally, the break-even value is markedly lower (26%) in our study compared to the thermochemical approach, which will be a considerable controlling variable in technology commercialization.

4. Conclusions

A brief description was given of potential routes to produce methanol from CO_2 and H_2 using solar energy. Photocatalytic water splitting is a low-cost H_2 source, but the simultaneous production of H_2 and O_2 in a closed environment introduces significant explosion hazards. In our study, we presented a new approach that utilized captured CO_2 as a diluent to recover the H_2 safely from the photocatalytic reactor and as a raw material to produce methanol. The process consisted of four plants, namely: a CO_2 capture unit (Plant 1), photocatalytic reactor (Plant 2), membrane separation plant (Plant 3), and methanol synthesis loop based on direct CO_2 hydrogenation (Plant 4). Plants 1 & 2 were not simulated in this work, but rather their associated costs were presented as purchasing raw materials. Process development and techno-economic analyses were carried on the separation and synthesis plants.

Two membrane layouts were optimized to recover both H_2 and CO_2 at a ratio of 3:1 while maintaining both safety and process constraints. A membrane cascade configuration (Layout (b)) comprised of three membrane units and one recycle stream, showed the lowest specific cost for H_2 recovery (\$3.08/kg$_{H_2\ recovered}$). The product stream from this plant was fed to the methanol plant. A reaction scheme based on a Cu/ZnO/Al_2O_3 catalyst was used to convert H_2

Table 9: Comparison between this study and methanol production via thermochemical conversion proposed by Kim, *et al.* [19]

	This Study	Kim and Coworkers
Technology used	Direct CO_2 hydrogenation using H_2 from photocatalytic water splitting	Methanol from CO_2 and water using thermochemical conversion of CO_2 to CO
Raw materials	Purchased CO_2 and H_2	Purchased CO_2 and de-ionized water
Economic Parameters		
- Project lifetime (T)	30 years	30 years
- Interest rate (i)	8%	8%
- CO_2 price	\$0.035/kg	\$0.035/kg
- Electricity Price	\$0.06/kWh	\$0.06/kWh
Results		
- Methanol yields	0.67 kg_{MeOH} / kg_{CO_2}	0.75 kg_{MeOH} / kg_{CO_2}
- Methanol purity	99.37%wt	99.06%wt
- Break-even value	\$0.94/kg_{MeOH}	\$1.22/kg_{MeOH}
- Net energy efficiency	28.18%	24.30%
- Gross energy efficiency	6.25%	10.40%

and CO_2 directly to methanol. The overall economics of the process revealed a high production cost compared to the conventional methanol production from natural gas. Given the base case parameters, a break-even value of methanol of \$1.30/kg_{MeOH} is needed to become economically attractive. Among the four plants, the separation plant has the highest negative cash flow effect with more than 49% of the total operating cost. Economic sensitivity analyses were carried for three parameters, namely: plant lifetime, electricity price and CO_2 capture price. The cost of the CO_2 raw materials has the highest impact. A 13.1% reduction on the break-even value was achieved when the capture price drop from 0.035 to 0.020 \$/kg_{CO_2}. Moreover, the best case scenario, where the plant is operated for 30 years with 0.060 \$/kWh electricity price and 0.020 \$/kg_{CO_2}, showed a 26.5% reduction in cost (\$0.96/kg_{MeOH}) compared to the base case. As for the energy efficiency, the proposed process demonstrated a 28.18% net energy efficiency. However, this value dropped to 6.25% for the gross energy efficiency when the total solar energy is accounted for (given an STH of 10%).

The proposed process shows promise for both production cost and energy efficiency as an alternative for renewably driven methanol production, but it is clear that further research and development is needed. These improvement techniques include developing higher performance membrane materials, lower CO_2 capture prices, enhanced STH efficiency of the photocatalysts, and systematic process integration between the plants. Moreover, variable ratios of H_2 and CO_2, leaving the separation plant, should be explored in order to enhance methanol production. This case study clearly identifies the potential of integrated CCR and renewable feedstock production to produce value-added commodity chemicals and highlights the opportunities for the scientific community to advance the field.

Acknowledgement

The authors would like to thank SABIC for funding the PhD work of Sari Alsayegh (main author).

List of symbols

α	equipment cost exponent factor	-
$A_{T,i}$	present value of annuity factor	year
BC	bare module cost	$
C_0	equipment reference cost	$
CC	capital cost	$
EAC	equivalent annual cost	$/year
i	interest rate	%
MF	module factor	-
MPF	material and pressure factor	-
NPV	net present value	$
OC	operating cost	$/year
R_t	net cash flow in year t	$
S	calculated equipment size	-
S_0	equipment reference size	equipment unit
SC	specific cost	$/kg
T	plant lifetime	year
UF	update factor	-

List of abbreviations

CCR	carbon dioxide capture and reuse	
DOE	U.S. department of energy	
GAMS	general algebraic modeling system	
HHV	high heating value	MJ/kmol
LFL	lower flammability limit	%
NLP	non-linear programming	
STH	solar to hydrogen	%
TMP_{max}	max. trans-membrane pressure	bar
UFL	upper flammability limit	%

Supporting Information

The supporting information document provide the followings:

- Detailed information on process equipment specifications (required for performance and economic calculations)
- Detailed process streams information for Plant (4)
- Energy analysis calculations for the overall process

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Methanol production via direct carbon dioxide hydrogenation using hydrogen from photocatalytic water splitting: process development and techno-economic analysis

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HIGHLIGHTS:

- Optimized H_2 recovery from water splitting with flammability constraints.
- Comparable production cost with other alternative routes for renewable methanol.
- High influence of CO_2 capture cost on the process economics.